

# Nguyen Khac Dong Quan

## Backend Developer Intern

✉ [quannguyen081204@gmail.com](mailto:quannguyen081204@gmail.com) | 📞 (+84) 0988 33 9801 | 📍 Ho Chi Minh City, Vietnam

🌐 [tmdkhac12.github.io](https://tmdkhac12.github.io) | 🐙 [github.com/tmdkhac12](https://github.com/tmdkhac12) | 💼 [linkedin.com/in/tmdkhac12](https://linkedin.com/in/tmdkhac12)

### OBJECTIVE

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[Position] with experience developing [type of systems/projects] using [core stack]. Short-term goal is to improve [specific technical area] and transition into a [target role] within [timeframe]. Seeking an opportunity at [Company Name] to apply technical skills, learn from experienced developers, and contribute effectively to real-world software projects.

### SKILLS

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**Programming Languages:** Java, JavaScript (Node.js)

**Core Concepts:** RESTful APIs, JWT, OAuth2, Socket.IO, Data Structures & Algorithms

**Backend Frameworks:** Spring Boot, Spring Security, Spring Data JPA, Express.js, Spring AI

**Databases & Persistence:** MySQL, MariaDB, Qdrant (Vector Database), Hibernate, MapStruct

**Software Architecture:** Layered Architecture, Modular Monolith, Monolithic Architecture

**Tools & AI Productivity:** Docker, Git, Linux, Github Copilot, Gemini CLI, Context/Token Management, Prompt Engineering

### PROJECTS

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**Perfume E-commerce** | Team size: 1 | 🐙 [GitHub Repository](#)

February 2026 – May 2026

*Technologies:* Java 21, Spring Boot 3, Spring AI, Qdrant (Vector DB), OAuth2/JWT, Docker, VNPay

- Built a **RAG-based AI consultant** with **Spring AI** and **Qdrant**, reducing manual support overhead by **70%** through 24/7 automated expert-level perfume advice.
- Integrated **VNPay v2.1** to provide a **seamless and secure cashless payment** experience, including a **Repay flow** to ensure successful transactions even after session timeouts.
- Integrated **Cloudinary** for automated asset management, offloading server storage and ensuring fast image delivery via CDN-based transformations.
- Designed a **modular monolith** architecture, accelerating development cycles for new features by **30%** through decoupled domains and MapStruct mapping.
- Optimized persistence with **EntityGraph** and **BatchSize**, reducing database queries **from N to 1-2** and improving response times by **90%** for heavy product listings.
- Architected **JWT + OAuth2 account linking** with SEO-friendly public access, streamlining the authentication flow and increasing user conversion rates.
- Built **async email notifications** via Spring Boot Mail, replacing manual invoicing labor and ensuring **100%** delivery of instant order confirmations.

**POS** | Team size: 1 | 🐙 [GitHub Repository](#)

May 2025 – August 2025

*Technologies:* Node.js, Express.js, MySQL, Socket.IO, EJS, RESTful APIs, express-session, bcrypt, Docker

- Engineered a **multi-directional real-time** system using **Socket.IO** with **Rooms/Namespaces**, isolating data by table/role and reducing **turnover time by 20%**.
- Architected instant event flows between **Customers** (order status), **Kitchen** (new orders), and **Staff** (support/payment requests) without page reloads.
- Designed a robust **order state machine** (Pending → Cooking → Served), maintaining strict data consistency across 3 distinct user roles.
- Ensured **data integrity** under high concurrency using **MySQL transactions**, preventing duplicate billing or lost orders during peak restaurant hours.
- Implemented a **hybrid SSR/API** architecture, providing fast initial loads for administrative dashboards while supporting dynamic client-side updates.
- Secured sensitive financial records through **RBAC middleware** and **Bcrypt** hashing, ensuring **0%** unauthorized access to revenue data.
- Developed an automated **reporting engine** using SQL aggregations, providing instant daily revenue analytics and customer loyalty tracking.

EXPERIENCE

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| <b>Nhat Minh Education Center</b><br><i>Math Tutor</i>                                                                                                                                                                                                                                                                                                                                                 | Feb 2025 – Aug 2025<br><i>Ho Chi Minh City, Vietnam</i> |
| <ul style="list-style-type: none"><li>Successfully mentored <b>10+ students</b>, significantly boosting academic results: <b>3 students jumped from failing grades (3-4) to excellence (9+)</b>, while others improved from 5-6 to 8+.</li><li>Tailored teaching methods to individual learning styles, demonstrating strong <b>patience, communication, and result-oriented mentorship</b>.</li></ul> |                                                         |
| <b>Freelance Web Development</b><br><i>Frontend Developer</i>                                                                                                                                                                                                                                                                                                                                          | Jan 2024 – Present<br><i>Ho Chi Minh City, Vietnam</i>  |
| <ul style="list-style-type: none"><li>Developed <b>5+ customized static web solutions</b> using <b>HTML, CSS, and JavaScript</b> based on specific academic project requirements for university students.</li><li>Ensured <b>100% adherence to technical specifications</b> and design handoffs, delivering clean and maintainable code for various educational modules.</li></ul>                     |                                                         |
| <b>Computer Repair Services</b><br><i>Freelance Technician</i>                                                                                                                                                                                                                                                                                                                                         | Jan 2023 – Present<br><i>Ho Chi Minh City, Vietnam</i>  |
| <ul style="list-style-type: none"><li>Resolved <b>20+ hardware and software cases</b> with high client satisfaction; performed Windows installations, RAM upgrades, and system troubleshooting.</li><li>Built a reputation for <b>reliable technical support</b> and quick problem diagnosis for individual clients in the local area.</li></ul>                                                       |                                                         |

EDUCATION

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| <b>Sai Gon University</b><br><i>Bachelor of Engineering in Information Technology</i> | Sep 2022 – Mar 2027<br><i>Current GPA: 7.8/10</i> |
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CERTIFICATES

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| <b>Aptis ESOL B2</b><br><i>English Proficiency Certificate</i> | Mar 2026<br><i>British Council (BC)</i> |
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HOBBIES

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| <b>Reading Books</b>                                                                                             |  |
| Regularly read books about psychology and communication to improve interpersonal and analytical thinking skills. |  |
| <b>Playing Piano</b>                                                                                             |  |
| Practice piano during free time as a hobby to relax and improve focus and discipline.                            |  |